

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Artificial Intelligence Application Development and Jira

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PRACTICAL 7

AIM :- Machine Learning

- 1) Using any small dataset (e.g., Iris or a CSV dataset), write a Python program to demonstrate the basic machine learning workflow.
 - a. Load dataset using Pandas.
 - b. Perform basic preprocessing (handling missing values or scaling).
 - c. Split the dataset into training and testing sets.
 - d. Train a simple classifier.
 - e. Display training and testing accuracy.

CODE :-

```
import pandas as pd

from sklearn.preprocessing import LabelEncoder, MinMaxScaler, StandardScaler

df = pd.read_csv("iris - iris - iris.csv")

print("\n===== 1. LOAD THE DATASET =====")

print(df)

print("\nSaqlain Khan T013")

print("\n===== 2. DISPLAY THE DATASET =====")

print(df)

print("\nSaqlain Khan T013")

print("\n===== 3. CHECK DATASET INFORMATION =====")

df.info()

print("\nSaqlain Khan T013")
```

```

print("\n===== 4. DISPLAY DATASET SHAPE =====")

print(df.shape)

print("\nSaqlain Khan T013")

print("\n===== 5. DISPLAY COLUMN NAMES =====")

print(df.columns)

print("\nSaqlain Khan T013")

print("\n===== 6. DISPLAY SUMMARY STATISTICS =====")

print(df.describe())

print("\nSaqlain Khan T013")

print("\n===== 7. CHECK DATA TYPES =====")

print(df.dtypes)

print("\nSaqlain Khan T013")

print("\n===== 8. CHECK MISSING VALUES =====")

print(df.isnull().sum())

print("\nSaqlain Khan T013")

print("\n===== 9. HANDLE MISSING VALUES =====")

df.fillna(df.mean(numeric_only=True), inplace=True)

print(df.isnull().sum())

print("\nSaqlain Khan T013")

print("\n===== 10. REMOVE DUPLICATE RECORDS =====")

df.drop_duplicates(inplace=True)

print(df)

print("\nSaqlain Khan T013")

print("\n===== 11. ENCODE CATEGORICAL DATA =====")

encoder = LabelEncoder()

df["species"] = encoder.fit_transform(df["species"])

```

```

print(df)

print("\nSaqlain Khan T013")

print("\n===== 12. RENAME COLUMNS =====")

df.rename(columns={

"sepal_length": "Sepal_Length",

"sepal_width": "Sepal_Width",

"petal_length": "Petal_Length",

"petal_width": "Petal_Width",

"species": "Species"

}, inplace=True)

print(df)

print("\nSaqlain Khan T013")

print("\n===== 13. DROP UNNECESSARY COLUMNS =====")

df.drop(columns=[], inplace=True)

print(df)

print("\nSaqlain Khan T013")

print("\n===== 14. SELECT FEATURES =====")

X = df.drop("Species", axis=1)

print(X)

print("\nSaqlain Khan T013")

print("\n===== 15. CREATE NEW FEATURE =====")

df["Total_Length"] = df["Sepal_Length"] + df["Petal_Length"]

print(df)

print("\nSaqlain Khan T013")

print("\n===== 16. CONVERT DATA TYPES =====")

df["Species"] = df["Species"].astype(int)

```

```

print(df.dtypes)

print("\nSaqlain Khan T013")

print("\n===== 17. NORMALIZE DATA (MIN-MAX SCALING) =====")

scaler = MinMaxScaler()

normalized_data = scaler.fit_transform(X)

print(normalized_data)

print("\nSaqlain Khan T013")

print("\n===== 18. STANDARDIZE DATA (STANDARD SCALING) =====")

standard_scaler = StandardScaler()

standardized_data = standard_scaler.fit_transform(X)

print(standardized_data)

print("\nSaqlain Khan T013")

print("\n===== 19. SPLIT FEATURES AND TARGET VARIABLE =====")

X = df.drop(["Species"], axis=1)

y = df["Species"]

print("Features (X):")

print(X)

print("\nTarget (y):")

print(y)

print("\nSaqlain Khan T013")

```

OUTPUT :-

- Load and Display the dataset

```

===== 2. DISPLAY THE DATASET =====
   sepal_length  sepal_width  petal_length  petal_width  species
0             5.1           3.5           1.4           0.2    setosa
1             4.9           3.0           1.4           0.2    setosa
2             4.7           3.2           1.3           0.2    setosa
3             4.6           3.1           1.5           0.2    setosa
4             5.0           3.6           1.4           0.2    setosa
..           ...           ...           ...           ...     ...
145            6.7           3.0           5.2           2.3  virginica
146            6.3           2.5           5.0           1.9  virginica
147            6.5           3.0           5.2           2.0  virginica
148            6.2           3.4           5.4           2.3  virginica
149            5.9           3.0           5.1           1.8  virginica

```

[150 rows x 5 columns]

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- Check dataset information

```

===== 3. CHECK DATASET INFORMATION =====
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
 #   Column          Non-Null Count  Dtype
---  ---
 0   sepal_length    150 non-null   float64
 1   sepal_width     150 non-null   float64
 2   petal_length    150 non-null   float64
 3   petal_width     150 non-null   float64
 4   species         150 non-null   object
dtypes: float64(4), object(1)
memory usage: 6.0+ KB

```

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- Display dataset shape

```

===== 4. DISPLAY DATASET SHAPE =====
(150, 5)

```

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- Display column names

```

===== 5. DISPLAY COLUMN NAMES =====
Index(['sepal_length', 'sepal_width', 'petal_length', 'petal_width',
       'species'],
      dtype='object')

```

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- Display summary statistics

```

===== 6. DISPLAY SUMMARY STATISTICS =====
      sepal_length  sepal_width  petal_length  petal_width
count    150.000000    150.000000    150.000000    150.000000
mean       5.843333     3.054000     3.758667     1.198667
std        0.828066     0.433594     1.764420     0.763161
min         4.300000     2.000000     1.000000     0.100000
25%         5.100000     2.800000     1.600000     0.300000
50%         5.800000     3.000000     4.350000     1.300000
75%         6.400000     3.300000     5.100000     1.800000
max         7.900000     4.400000     6.900000     2.500000

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```

- Check data types

```

===== 7. CHECK DATA TYPES =====
sepal_length    float64
sepal_width     float64
petal_length    float64
petal_width     float64
species         object
dtype: object

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```

- Check missing values

```

===== 8. CHECK MISSING VALUES =====
sepal_length    0
sepal_width     0
petal_length    0
petal_width     0
species         0
dtype: int64

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```

- Handle missing values

```

===== 9. HANDLE MISSING VALUES =====
sepal_length    0
sepal_width     0
petal_length    0
petal_width     0
species         0
dtype: int64

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```

- Remove duplicate records

```

===== 10. REMOVE DUPLICATE RECORDS =====
      sepal_length  sepal_width  petal_length  petal_width  species
0           5.1         3.5         1.4         0.2     setosa
1           4.9         3.0         1.4         0.2     setosa
2           4.7         3.2         1.3         0.2     setosa
3           4.6         3.1         1.5         0.2     setosa
4           5.0         3.6         1.4         0.2     setosa
..         ...         ...         ...         ...         ...
145          6.7         3.0         5.2         2.3  virginica
146          6.3         2.5         5.0         1.9  virginica
147          6.5         3.0         5.2         2.0  virginica
148          6.2         3.4         5.4         2.3  virginica
149          5.9         3.0         5.1         1.8  virginica

[147 rows x 5 columns]

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```

- Encode categorical data

```

===== 11. ENCODE CATEGORICAL DATA =====
      sepal_length  sepal_width  petal_length  petal_width  species
0           5.1         3.5         1.4         0.2         0
1           4.9         3.0         1.4         0.2         0
2           4.7         3.2         1.3         0.2         0
3           4.6         3.1         1.5         0.2         0
4           5.0         3.6         1.4         0.2         0
..         ...         ...         ...         ...         ...
145          6.7         3.0         5.2         2.3         2
146          6.3         2.5         5.0         1.9         2
147          6.5         3.0         5.2         2.0         2
148          6.2         3.4         5.4         2.3         2
149          5.9         3.0         5.1         1.8         2

[147 rows x 5 columns]

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```

- Rename columns

```

===== 12. RENAME COLUMNS =====
      Sepal_Length  Sepal_Width  Petal_Length  Petal_Width  Species
0           5.1         3.5         1.4         0.2         0
1           4.9         3.0         1.4         0.2         0
2           4.7         3.2         1.3         0.2         0
3           4.6         3.1         1.5         0.2         0
4           5.0         3.6         1.4         0.2         0
..         ...         ...         ...         ...         ...
145          6.7         3.0         5.2         2.3         2
146          6.3         2.5         5.0         1.9         2
147          6.5         3.0         5.2         2.0         2
148          6.2         3.4         5.4         2.3         2
149          5.9         3.0         5.1         1.8         2

[147 rows x 5 columns]

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```

- Drop unnecessary columns

```

===== 13. DROP UNNECESSARY COLUMNS =====
      Sepal_Length  Sepal_Width  Petal_Length  Petal_Width  Species
0                5.1          3.5          1.4          0.2          0
1                4.9          3.0          1.4          0.2          0
2                4.7          3.2          1.3          0.2          0
3                4.6          3.1          1.5          0.2          0
4                5.0          3.6          1.4          0.2          0
..                ...          ...          ...          ...          ...
145              6.7          3.0          5.2          2.3          2
146              6.3          2.5          5.0          1.9          2
147              6.5          3.0          5.2          2.0          2
148              6.2          3.4          5.4          2.3          2
149              5.9          3.0          5.1          1.8          2

[147 rows x 5 columns]

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```

- Select features

```

===== 14. SELECT FEATURES =====
      Sepal_Length  Sepal_Width  Petal_Length  Petal_Width
0                5.1          3.5          1.4          0.2
1                4.9          3.0          1.4          0.2
2                4.7          3.2          1.3          0.2
3                4.6          3.1          1.5          0.2
4                5.0          3.6          1.4          0.2
..                ...          ...          ...          ...
145              6.7          3.0          5.2          2.3
146              6.3          2.5          5.0          1.9
147              6.5          3.0          5.2          2.0
148              6.2          3.4          5.4          2.3
149              5.9          3.0          5.1          1.8

[147 rows x 4 columns]

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```

- Create new features

```

===== 15. CREATE NEW FEATURE =====
      Sepal_Length  Sepal_Width  Petal_Length  Petal_Width  Species  \
0                5.1          3.5          1.4          0.2          0
1                4.9          3.0          1.4          0.2          0
2                4.7          3.2          1.3          0.2          0
3                4.6          3.1          1.5          0.2          0
4                5.0          3.6          1.4          0.2          0
..                ...          ...          ...          ...          ...
145              6.7          3.0          5.2          2.3          2
146              6.3          2.5          5.0          1.9          2
147              6.5          3.0          5.2          2.0          2
148              6.2          3.4          5.4          2.3          2
149              5.9          3.0          5.1          1.8          2

      Total_Length
0                6.5
1                6.3
2                6.0
3                6.1
4                6.4
..                ...
145              11.9
146              11.3
147              11.7
148              11.6
149              11.0

[147 rows x 6 columns]

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```

- Convert data types


```

===== 16. CONVERT DATA TYPES =====
Sepal_Length    float64
Sepal_Width     float64
Petal_Length    float64
Petal_Width     float64
Species         int64
Total_Length    float64
dtype: object

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```

- Normalize data

```

===== 17. NORMALIZE DATA (MIN-MAX SCALING) =====
[[0.22222222 0.625      0.06779661 0.04166667]
 [0.16666667 0.41666667 0.06779661 0.04166667]
 [0.11111111 0.5       0.05084746 0.04166667]
 [0.08333333 0.45833333 0.08474576 0.04166667]
 [0.19444444 0.66666667 0.06779661 0.04166667]
 [0.30555556 0.79166667 0.11864407 0.125      ]
 [0.08333333 0.58333333 0.06779661 0.08333333]
 [0.19444444 0.58333333 0.08474576 0.04166667]
 [0.02777778 0.375      0.06779661 0.04166667]

```

- Standardize data

```

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===== 18. STANDARDIZE DATA (STANDARD SCALING) =====
[[-0.91550949  1.0199714 -1.35773692 -1.33570001]
 [-1.15755966 -0.12808217 -1.35773692 -1.33570001]
 [-1.39960983  0.33113926 -1.41477817 -1.33570001]
 [-1.52063491  0.10152855 -1.30069567 -1.33570001]
 [-1.03653457  1.24958212 -1.35773692 -1.33570001]
 [-0.55243423  1.93841426 -1.18661318 -1.07090176]
 [-1.52063491  0.79036069 -1.35773692 -1.20330089]
 [-1.03653457  0.79036069 -1.30069567 -1.33570001]

```

- Split features and Target variable

```

===== 19. SPLIT FEATURES AND TARGET VARIABLE =====
Features (X):
   Sepal_Length  Sepal_Width  Petal_Length  Petal_Width  Total_Length
0             5.1           3.5           1.4           0.2           6.5
1             4.9           3.0           1.4           0.2           6.3
2             4.7           3.2           1.3           0.2           6.0
3             4.6           3.1           1.5           0.2           6.1
4             5.0           3.6           1.4           0.2           6.4
..            ...           ...           ...           ...           ...
145            6.7           3.0           5.2           2.3          11.9
146            6.3           2.5           5.0           1.9          11.3
147            6.5           3.0           5.2           2.0          11.7
148            6.2           3.4           5.4           2.3          11.6
149            5.9           3.0           5.1           1.8          11.0

[147 rows x 5 columns]

```

2) Different Dataset

CODE :-

```
import pandas as pd

from sklearn.preprocessing import LabelEncoder, MinMaxScaler, StandardScaler

df = pd.read_csv("supermarket.csv")

print("\n===== 1. LOAD THE DATASET =====")

print(df)

print("\nSaqlain Khan T013")

print("\n===== 2. DISPLAY THE DATASET =====")

print(df)

print("\nSaqlain Khan T013")

print("\n===== 3. CHECK DATASET INFORMATION =====")

df.info()

print("\nSaqlain Khan T013")

print("\n===== 4. DISPLAY DATASET SHAPE =====")

print(df.shape)

print("\nSaqlain Khan T013")

print("\n===== 5. DISPLAY COLUMN NAMES =====")

print(df.columns)

print("\nSaqlain Khan T013")

print("\n===== 6. DISPLAY SUMMARY STATISTICS =====")

print(df.describe())

print("\nSaqlain Khan T013")

print("\n===== 7. CHECK DATA TYPES =====")

print(df.dtypes)

print("\nSaqlain Khan T013")

print("\n===== 8. CHECK MISSING VALUES =====")

print(df.isnull().sum())

print("\nSaqlain Khan T013")

print("\n===== 9. HANDLE MISSING VALUES =====")
```

```

df.fillna(df.select_dtypes(include="number").mean(), inplace=True)

print(df.isnull().sum())

print("\nSaqlain Khan T013")

print("\n===== 10. REMOVE DUPLICATE RECORDS =====")

df.drop_duplicates(inplace=True)

print(df.shape)

print("\nSaqlain Khan T013")

print("\n===== 11. ENCODE CATEGORICAL DATA =====")

encoder = LabelEncoder()

categorical_columns = [

    "Branch",

    "City",

    "Customer type",

    "Gender",

    "Product line",

    "Payment"

]

for col in categorical_columns:

    df[col] = encoder.fit_transform(df[col])

print(df.head())

print("\nSaqlain Khan T013")

print("\n===== 12. RENAME COLUMNS =====")

df.rename(columns={

    "Invoice ID": "Invoice_ID",

    "Customer type": "Customer_Type",

    "Product line": "Product_Line",

    "Unit price": "Unit_Price",

    "Tax 5%": "Tax_5",

    "gross margin percentage": "Gross_Margin_Percentage",

    "gross income": "Gross_Income"

```

```

    }, inplace=True)

print(df.columns)

print("\nSaqlain Khan T013")

print("\n===== 13. DROP UNNECESSARY COLUMNS =====")

df.drop("Invoice_ID", axis=1, inplace=True)

print(df.head())

print("\nSaqlain Khan T013")

print("\n===== 14. SELECT FEATURES =====")

X = df.drop("Rating", axis=1)

print(X.head())

print("\nSaqlain Khan T013")

print("\n===== 15. CREATE NEW FEATURE =====")

df["Total_Amount"] = df["Unit_Price"] * df["Quantity"]

print(df.head())

print("\nSaqlain Khan T013")

print("\n===== 16. CONVERT DATA TYPES =====")

df["Quantity"] = df["Quantity"].astype(float)

print(df.dtypes)

print("\nSaqlain Khan T013")

print("\n===== 17. NORMALIZE DATA (MIN-MAX SCALING) =====")

numeric_columns = df.select_dtypes(include="number").columns

minmax = MinMaxScaler()

df[numeric_columns] = minmax.fit_transform(df[numeric_columns])

print(df.head())

print("\nSaqlain Khan T013")

print("\n===== 18. STANDARDIZE DATA (STANDARD SCALING) =====")

standard = StandardScaler()

df[numeric_columns] = standard.fit_transform(df[numeric_columns])

```

```

print(df.head())

print("\nSaqlain Khan T013")

print("\n===== 19. SPLIT FEATURES AND TARGET VARIABLE
=====")

X = df.drop("Rating", axis=1)

y = df["Rating"]

print("Features (X)")

print(X.head())

print("\nTarget (y)")

print(y.head())

print("\nSaqlain Khan T013")

```

OUTPUT :-

- Load and Display the dataset

```

===== 2. DISPLAY THE DATASET =====

```

	Invoice ID	Branch	City	Customer type	Gender \
0	750-67-8428	A	Yangon	Member	Female
1	226-31-3081	C	Naypyitaw	Normal	Female
2	631-41-3108	A	Yangon	Normal	Male
3	123-19-1176	A	Yangon	Member	Male
4	373-73-7910	A	Yangon	Normal	Male

- Check dataset information

```

===== 3. CHECK DATASET INFORMATION =====
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 17 columns):
#   Column              Non-Null Count  Dtype
---  ---
0   Invoice ID           1000 non-null   object
1   Branch              1000 non-null   object

```

- Display dataset shape

```

===== 4. DISPLAY DATASET SHAPE =====
(1000, 17)

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```

- Display column names

```

===== 5. DISPLAY COLUMN NAMES =====
Index(['Invoice ID', 'Branch', 'City', 'Customer type', 'Gender',
       'Product line', 'Unit price', 'Quantity', 'Tax 5%', 'Total', 'Date',
       'Time', 'Payment', 'cogs', 'gross margin percentage', 'gross income',
       'Rating'],
      dtype='object')

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```

- Display summary statistics

```
===== 6. DISPLAY SUMMARY STATISTICS =====
      Unit price    Quantity    Tax 5%    Total    cogs \
count  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000
mean    55.672130    5.510000    15.379369    322.966749    307.58738
std     26.494628    2.923431    11.708825    245.885335    234.17651
```

- Check data types

```
===== 7. CHECK DATA TYPES =====
Invoice ID      object
Branch          object
City            object
Customer type   object
```

- Check missing values

```
===== 8. CHECK MISSING VALUES =====
Invoice ID      0
Branch          0
City            0
Customer type   0
Gender          0
```

- Handle missing values

```
===== 9. HANDLE MISSING VALUES =====
Invoice ID      0
Branch          0
City            0
Customer type   0
Gender          0
```

- Remove duplicate records

```
===== 10. REMOVE DUPLICATE RECORDS =====
(1000, 17)

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```

- Encode categorical columns

```
===== 11. ENCODE CATEGORICAL DATA =====
      Invoice ID  Branch  City  Customer type  Gender  Product line  Unit price \
0  750-67-8428    0      2          0          0          3        74.69
1  226-31-3081    2      1          1          0          0        15.28
2  631-41-3108    0      2          1          1          4        46.33
3  123-19-1176    0      2          0          1          3        58.22
```

- Rename columns

```
===== 12. RENAME COLUMNS =====
Index(['Invoice_ID', 'Branch', 'City', 'Customer_Type', 'Gender',
      'Product_Line', 'Unit_Price', 'Quantity', 'Tax_5', 'Total', 'Date',
      'Time', 'Payment', 'cogs', 'Gross_Margin_Percentage', 'Gross_Income',
      'Rating'],
      dtype='object')
```

- Drop unnecessary columns

```
===== 13. DROP UNNECESSARY COLUMNS =====
   Branch  City  Customer_Type  Gender  Product_Line  Unit_Price  Quantity  \
0         0     2              0       0             3        74.69         7
1         2     1              1       0             0        15.28         5
2         0     2              1       1             4        46.33         7
3         0     2              0       1             3        58.22         8
4         0     2              1       1             5        86.31         7
```

- Select features

```
===== 14. SELECT FEATURES =====
   Branch  City  Customer_Type  Gender  Product_Line  Unit_Price  Quantity  \
0         0     2              0       0             3        74.69         7
1         2     1              1       0             0        15.28         5
2         0     2              1       1             4        46.33         7
3         0     2              0       1             3        58.22         8
4         0     2              1       1             5        86.31         7
```

- Create new features

```
===== 15. CREATE NEW FEATURE =====
   Branch  City  Customer_Type  Gender  Product_Line  Unit_Price  Quantity  \
0         0     2              0       0             3        74.69         7
1         2     1              1       0             0        15.28         5
2         0     2              1       1             4        46.33         7
3         0     2              0       1             3        58.22         8
```

- Convert data types

```
===== 16. CONVERT DATA TYPES =====
Branch                int64
City                  int64
Customer_Type         int64
Gender                 int64
Product_Line          int64
Unit_Price            float64
Quantity              float64
```

- Normalize data

```
===== 17. NORMALIZE DATA (MIN-MAX SCALING) =====
   Branch  City  Customer_Type  Gender  Product_Line  Unit_Price  Quantity  \
0     0.0    1.0              0.0     0.0           0.6     0.718847     0.666667
1     1.0    0.5              1.0     0.0           0.0     0.057855     0.444444
2     0.0    1.0              1.0     1.0           0.8     0.403316     0.666667
3     0.0    1.0              0.0     1.0           0.6     0.535603     0.777778
```

- Standardize data

```
===== 18. STANDARDIZE DATA (STANDARD SCALING) =====
   Branch  City  Customer_Type  Gender  Product_Line  Unit_Price  \
0 -1.208970  1.210174    -0.998002  -0.998002    0.319617    0.718160
1  1.238338 -0.009759     1.002002  -0.998002   -1.430109   -1.525303
2 -1.208970  1.210174     1.002002  1.002002    0.902859   -0.352781
3 -1.208970  1.210174    -0.998002  1.002002    0.319617    0.096214
```

- Split features

===== 19. SPLIT FEATURES AND TARGET VARIABLE =====

Features (X)

	Branch	City	Customer_Type	Gender	Product_Line	Unit_Price	\
0	-1.208970	1.210174	-0.998002	-0.998002	0.319617	0.718160	
1	1.238338	-0.009759	1.002002	-0.998002	-1.430109	-1.525303	

Feature	Iris Dataset	Supermarket Dataset
Dataset Type	Flower dataset	Sales transaction dataset
Number of Rows	150	1000
Number of Columns	5	17
Target Variable	Species	Rating (for practice)
Data Type	Mostly Numeric	Numeric + Categorical
Missing Values	Usually None	Usually None
Duplicate Records	Rare	May Exist
Encoding Needed	Species only	Branch, City, Gender, Payment, etc.
Scaling Needed	Yes	Yes
Feature Engineering	Total_Length	Total_Amount
Unnecessary Columns	None	Invoice_ID
Machine Learning Task	Classification	Classification / Regression
Complexity	Simple	Moderate
Best Used For	Learning ML Basics	Data Preprocessing & Sales Analysis

2) Using a suitable regression dataset, implement Linear Regression using scikit-Learn.

CODE :-

OUTPUT :-